

Smit Chaudhary

SOFTWARE ENGINEER · ALGORITHMS DEVELOPER · QUANTUM PHYSICIST

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Experience

Quantum Algorithms Researcher

Amsterdam, The Netherlands

PASQAL

Nov 2022 - Present

- Built Quantum Scientific ML (QSciML) framework with multi-backend support (**PyTorch**, **JAX**) and advanced differentiation modes, enabling novel neural architectures for PDE solving
- Delivered production **APIs** and **cloud SDKs** for aerospace industry partners, managing technical partnerships and translating research algorithms into industry-ready solutions
- Implemented **distributed ML infrastructure** and experiment tracking systems using **Dask** and **MLFlow**, reducing experiment turnaround time and enabling parallel hyperparameter optimization
- Developed dynamic growth-based training algorithms that adaptively increase model depth during training to mitigate overfitting, resulting in **research publication**

Quantum Computing Intern

California, USA

MENTEN AI

May 2022 - Sept 2022

- Built quantum **generative adversarial networks (QGANS)** in **JAX**, to solve discrete data generation tasks, implementing custom feature encoding schemes that enable learning from sparse datasets
- Designed experiment **validation framework** with testing across multiple initializations and dataset configurations, demonstrating model performance on both synthetic and physics-based optimization tasks
- Published **peer-reviewed research** on discrete generative modeling approaches that mitigate computational scalability limitations

Education

Technische Universiteit Delft

Delft, The Netherlands

MASTER OF SCIENCE IN APPLIED PHYSICS

August 2020 - July 2022

- **Thesis:** *Quantum Walk Based Algorithms for Qubit Mapping*
Built combinatorial optimization algorithms in Python using constraint satisfaction and greedy routing approaches
- **Research Project:** *Generative Adversarial Network Framework*
Developed custom ML framework with gradient computation engine from scratch using NumPy with modular OOP architecture
- **Relevant Coursework:** Machine Learning, Information Theory, Object-Oriented Programming (C++), Computational Physics

Indian Institute of Technology, Kanpur

Kanpur, India

BACHELOR OF SCIENCE IN PHYSICS

July 2016 - May 2020

- **Research Experience:** *Quantum Machine Learning algorithms research* at IISER Kolkata
- **Undergraduate Project:** *Bohmian Mechanics and Quantum Information*
- **Relevant Coursework:** Fundamentals of Computing, Probability and Statistics, Mathematical Methods, Electronics

Selected Publications

Towards a Scalable Discrete Quantum Generative Adversarial Neural Network

S. Chaudhary, P. Huembeli, I. MacCormack, T. L. Patti, J. Kossaifi, A. Galda

Quantum Circuit Training with Growth-Based Architectures

C. Duffy, S. Chaudhary, G. V. Velikova

Solving Fluid Dynamics Equations with Differentiable Quantum Circuits

S. Chaudhary, G. T. Balducci, O. Kyriienko, P. K. Barkoutsos, L. Cardarelli, A. A. Gentile

Skills

Programming

Python, C/C++, Rust, Julia

ML & Data Tools

PyTorch, JAX, NumPy, TensorFlow, MLFlow, Dask

Development Tools

Git, GitLab CI/CD, Slurm

Languages

English, Hindi, Gujarati